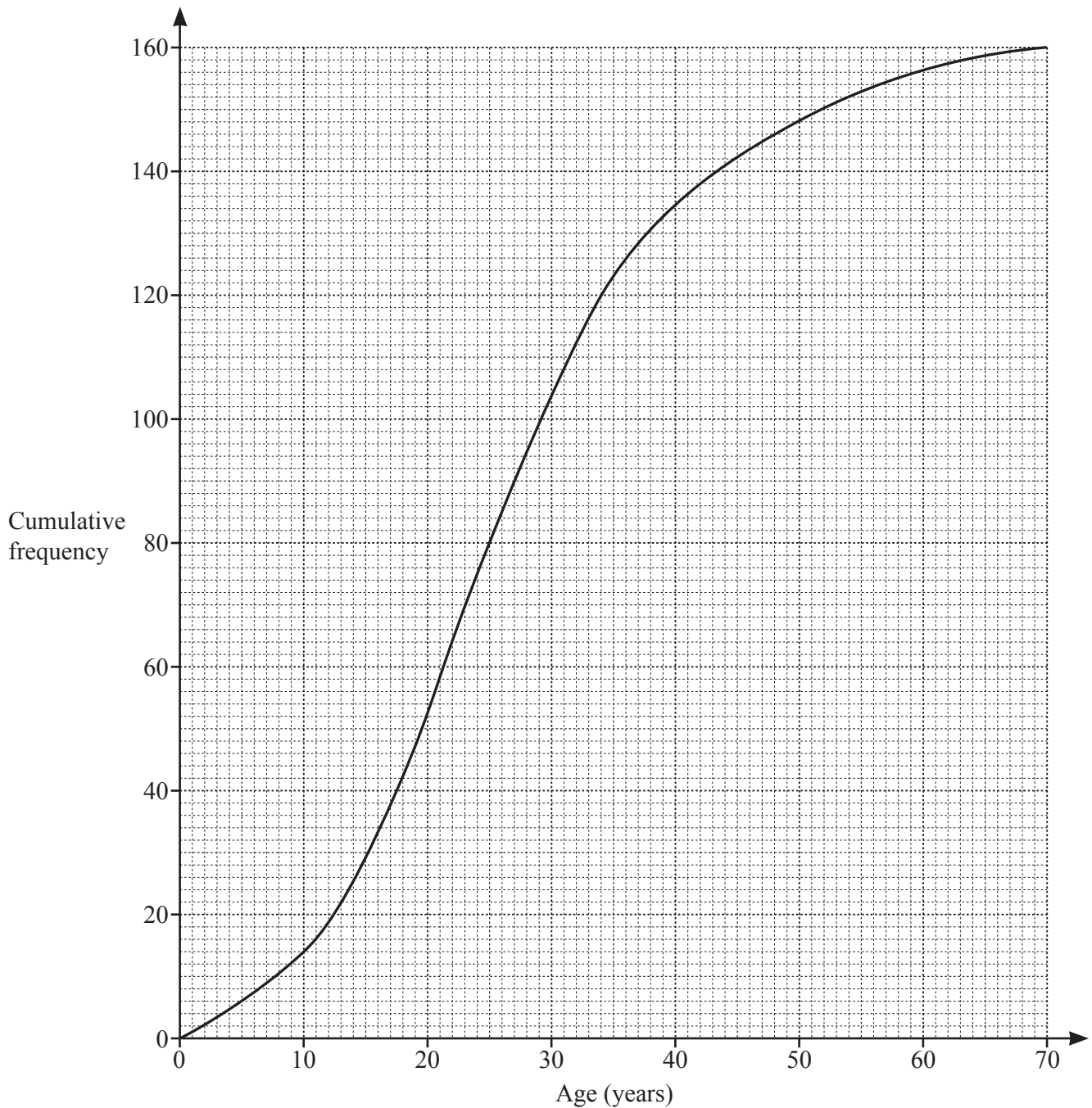


- 5 (a) There are 160 people in a village.
The cumulative frequency diagram shows information about their ages.



- (i) Find an estimate for

(a) the median age

..... [1]

(b) the lower quartile

..... [1]

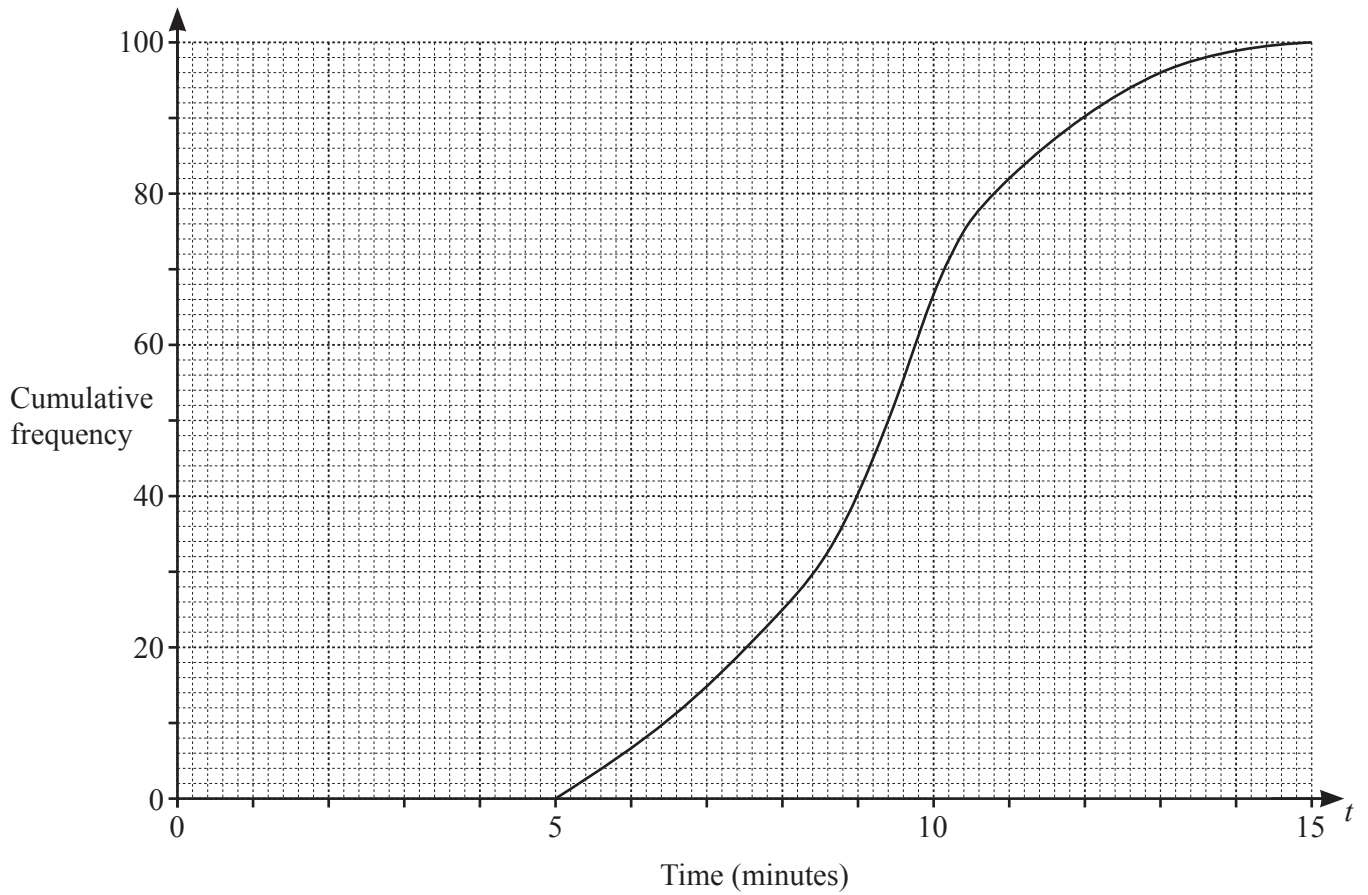
(c) the number of people who are 50 or more years of age

..... [2]

(d) the 65th percentile.

..... [2]

- 5 (a) 100 students each record the time, t minutes, taken to eat a pizza.
The cumulative frequency diagram shows the results.



Find an estimate of

- (i) the median,

..... min [1]

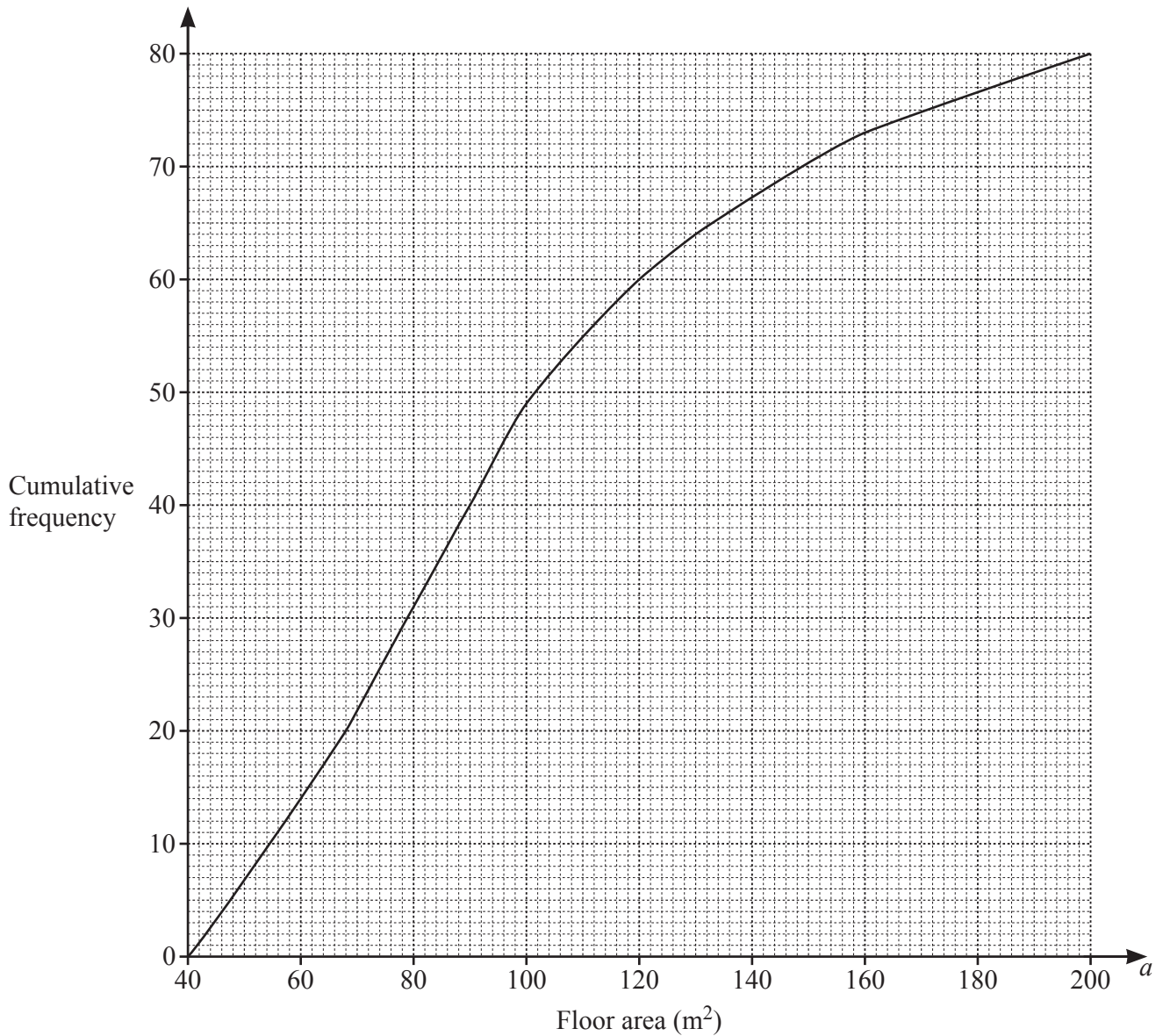
- (ii) the interquartile range,

..... min [2]

- (iii) the number of students taking more than 11 minutes to eat a pizza.

..... [2]

- 2 (a) The cumulative frequency diagram shows information about the floor area, $a \text{ m}^2$, of each of 80 houses.



Use the diagram to find an estimate of

- (i) the median, m^2 [1]
- (ii) the lower quartile, m^2 [1]
- (iii) the interquartile range, m^2 [1]
- (iv) the number of houses with a floor area greater than 120 m^2 .

..... [2]

- 3 (a) Zoe's test scores last term were 6 7 7 7 8 9 9 10 10.

Find

- (i) the range,

..... [1]

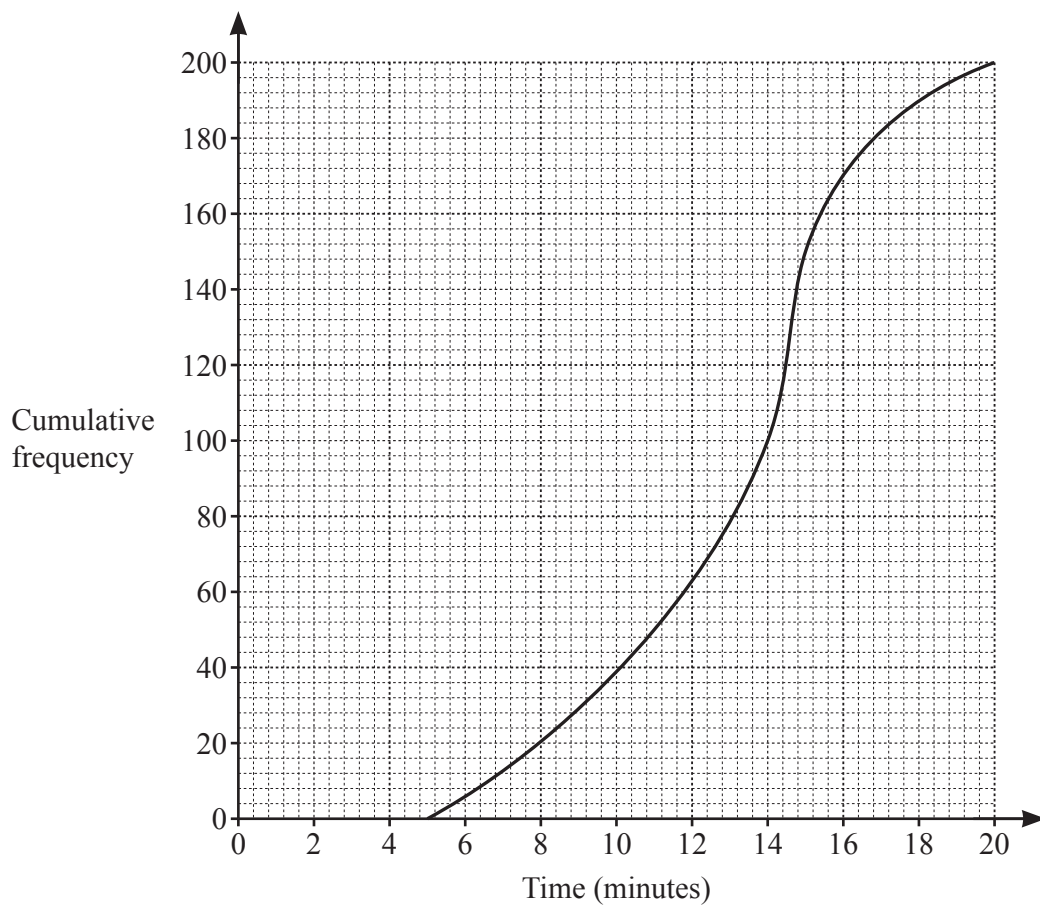
- (ii) the mode,

..... [1]

- (iii) the median.

..... [1]

- (b) The cumulative frequency diagram shows information about the time taken by each of 200 students to solve a problem.



Use the diagram to find an estimate of

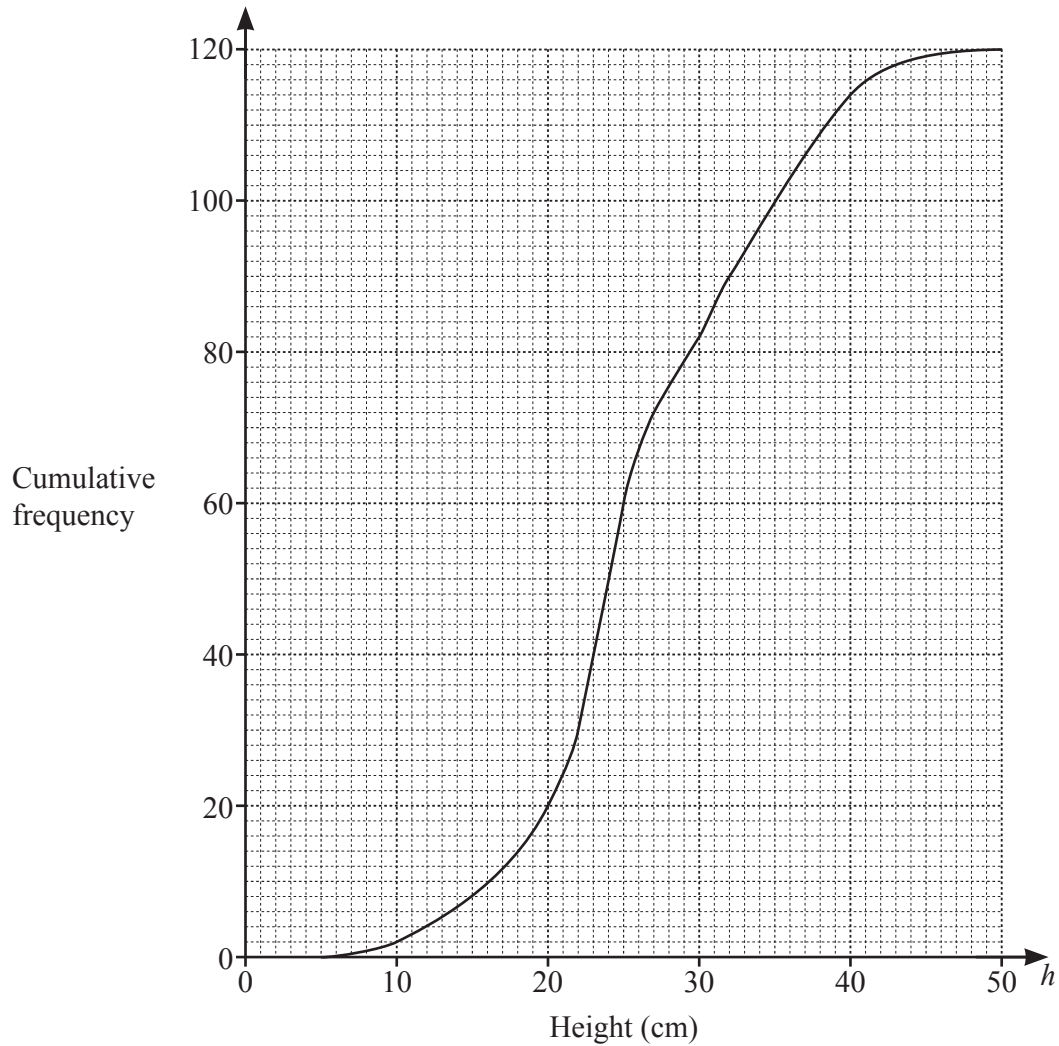
- (i) the median,

..... min [1]

- (ii) the interquartile range.

..... min [2]

- 4 The height, h cm, of each of 120 plants is measured.
The cumulative frequency diagram shows this information.



(a) Use the cumulative frequency diagram to find an estimate of

(i) the median,

..... cm [1]

(ii) the interquartile range,

..... cm [2]

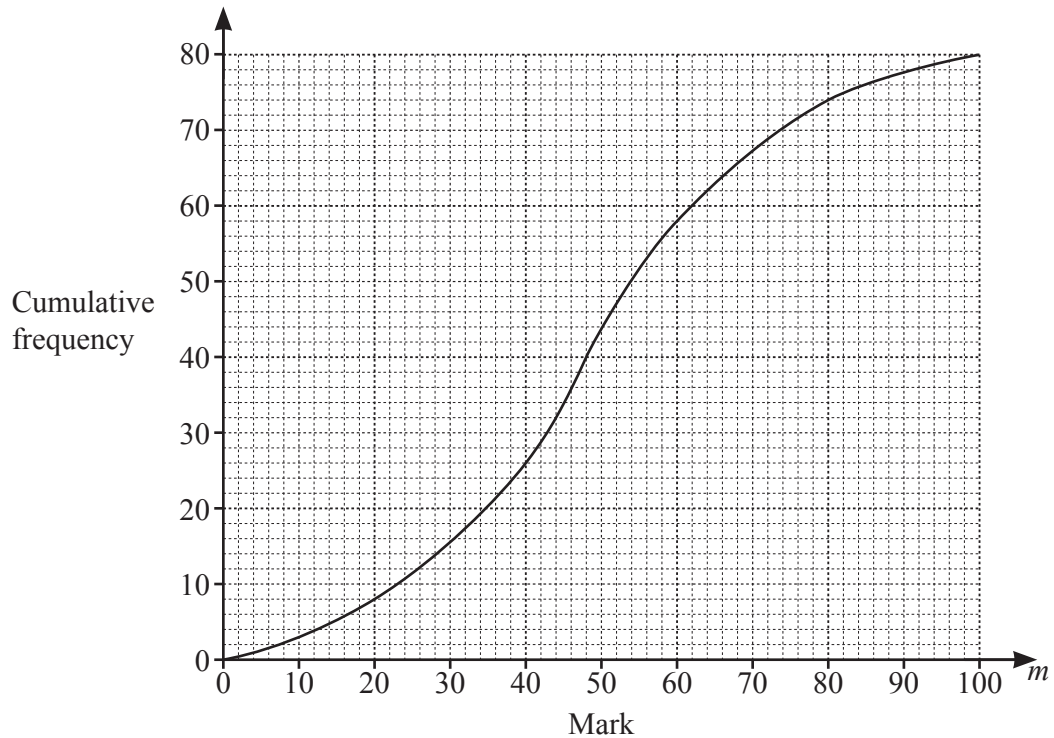
(iii) the 60th percentile,

..... cm [1]

(iv) the number of plants with a height greater than 40 cm.

..... [2]

- 9 (a) The cumulative frequency diagram shows the marks obtained by 80 students in a Maths test.



- (i) Use the diagram to find an estimate of the median.

..... [1]

- (ii) 60% of the students passed the test.

Use the diagram to find the number of marks needed to pass the test.

..... [2]

- (iii) Using the information on the diagram, complete the frequency table.

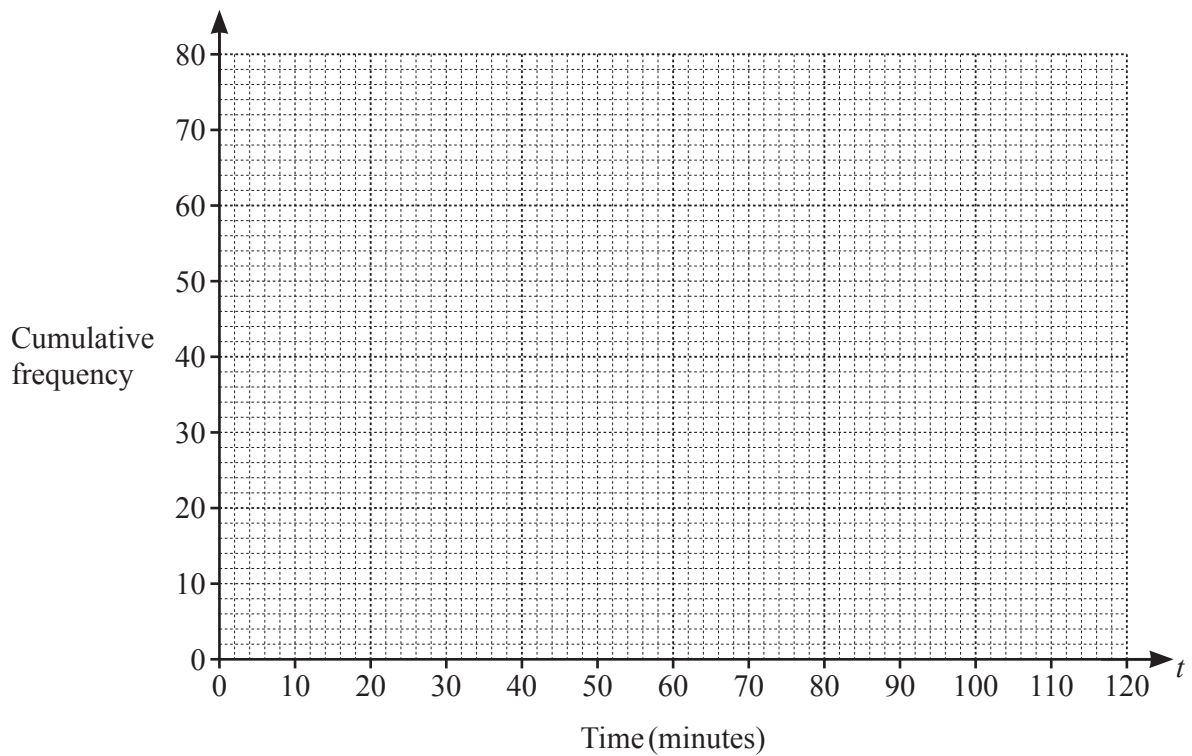
Mark (m)	$0 \leq m < 20$	$20 \leq m < 40$	$40 \leq m < 60$	$60 \leq m < 80$	$80 \leq m < 100$
Frequency	8				

[2]

- 2 (a) A group of 80 students each completed a task.
The table shows the time, t minutes, each student took to complete the task.

Time (t minutes)	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 80$	$80 < t \leq 100$	$100 < t \leq 120$
Frequency	10	20	34	12	4

- (i) On the grid, draw a cumulative frequency diagram to represent this information.



[3]

- (ii) Use your diagram to estimate

- (a) the median,

..... minutes [1]

- (b) the interquartile range.

..... minutes [2]

- (b) A group of 160 adults each completed the same task.
The table shows the number of errors made by each of these adults.

Number of errors	0	1	2	3	4	5
Frequency	24	30	50	32	16	8

- (i) Calculate the mean.

..... [2]

- (ii) One of the adults is selected at random.

Find the probability that this adult made more than 3 errors.

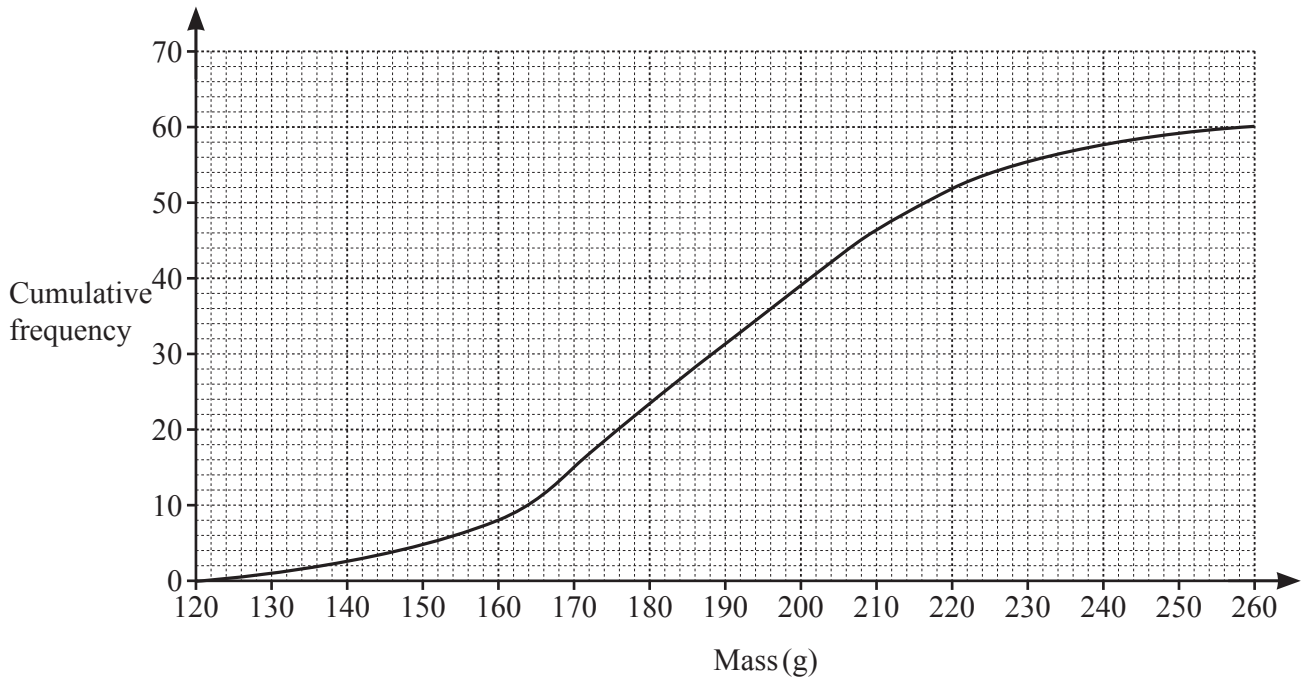
..... [1]

- (iii) Two of the adults are selected at random.

Find the probability that they each made exactly one error.

..... [2]

- 2 (a) The cumulative frequency diagram shows the masses, in grams, of 60 potatoes of variety A.



- (i) Use the diagram to estimate

(a) the median,

..... g [1]

(b) the interquartile range.

..... g [2]

- (ii) Potatoes with a mass greater than 220 g are sold as baking potatoes.

Find the percentage of the potatoes that are sold as baking potatoes.

..... % [3]

- (iii) The masses of 60 potatoes of variety B are also measured.
For variety B, the median is 175 g and the interquartile range is 30 g.

Kali wants to buy potatoes that are more consistent in mass.

Should she choose variety A or variety B?
Explain how you decide.

Variety because [1]

(b) The table shows the masses, m grams, of 120 potatoes of variety C.

Mass (m g)	$80 \leq m < 100$	$100 \leq m < 120$	$120 \leq m < 130$	$130 \leq m < 140$	$140 \leq m < 200$
Frequency	10	15	42	36	17

Calculate an estimate of the mean mass.

..... g [3]

- (c) A bag of potatoes has a mass of 2.5 kg, correct to the nearest 100 g.
 Bags of potatoes are packed into a box.
 The mass of the box is 600 g, correct to the nearest 10 g.

Calculate the upper bound of the total mass, in kilograms, of a box containing 10 of these bags of potatoes.

..... kg [3]

- 2 A company asked their employees how long they took to travel to work one day. The table summarises the times for 120 employees.

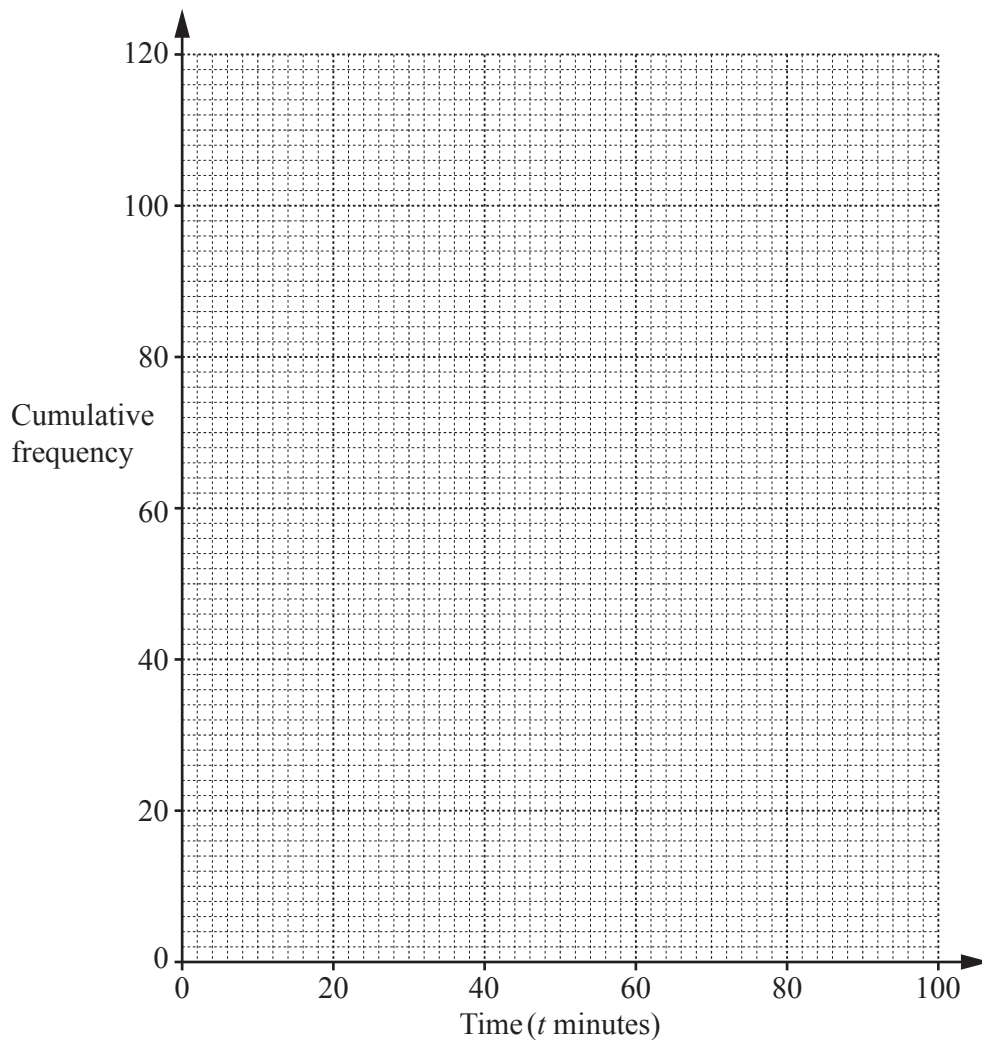
Time (t minutes)	$0 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 80$	$80 < t \leq 100$
Frequency	12	28	45	22	13

- (a) (i) Complete the cumulative frequency table below.

Time (t minutes)	$t \leq 0$	$t \leq 20$	$t \leq 40$	$t \leq 60$	$t \leq 80$	$t \leq 100$
Cumulative frequency	0					120

[1]

- (ii) On the grid, draw a smooth cumulative frequency curve to represent these results.



[2]

(b) Use your curve to estimate

(i) the median time,

Answer minutes [1]

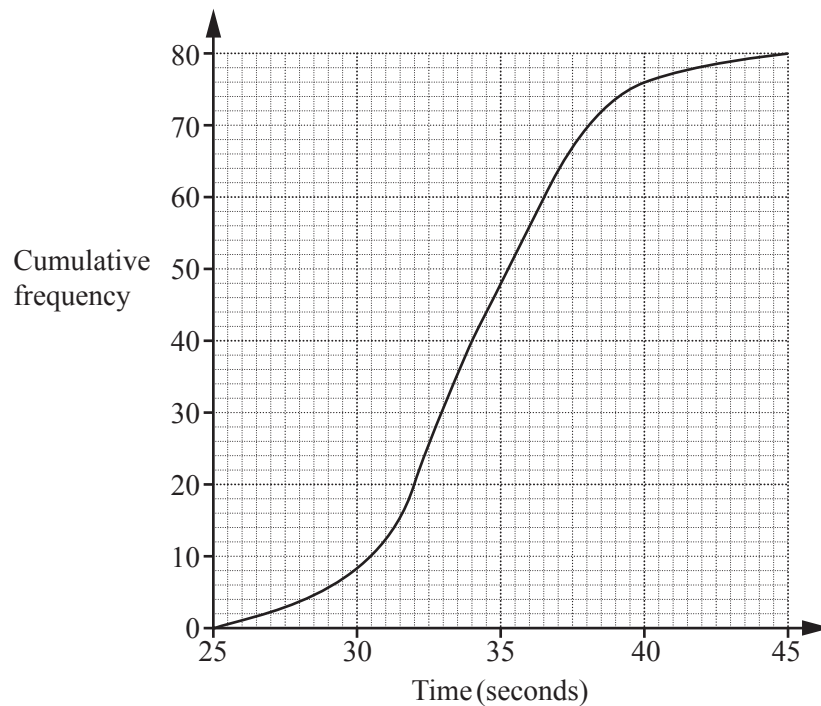
(ii) the interquartile range of the times.

Answer minutes [2]

(c) Calculate an estimate of the mean time taken for the employees to travel to work.

Answer minutes [3]

- (b) The cumulative frequency curve summarises the times taken by 80 boys to run 200 m.



- (i) Find the median time.

Answer s [1]

- (ii) Find the interquartile range.

Answer s [2]

- (iii) 60 girls also ran 200 m.

The girl who took the longest time ran 200 m in 40 seconds.
The girl who took the shortest time ran 200 m in 28 seconds.

The lower quartile for the boys and the girls is the same.
The interquartile range for the girls is 4 seconds.

Draw the cumulative frequency curve on the grid above.

[3]

- 11 80 people were each asked how much they spent on clothes last month. The results are summarised in the table below.

Amount spent (\$ c)	Frequency
$0 < c \leq 20$	3
$20 < c \leq 40$	8
$40 < c \leq 60$	14
$60 < c \leq 80$	21
$80 < c \leq 100$	18
$100 < c \leq 120$	9
$120 < c \leq 140$	5
$140 < c \leq 160$	2

- (a) Calculate an estimate of the mean amount spent on clothes last month.

Answer \$ [3]

- (b) Complete the cumulative frequency table below.

Amount spent (\$ c)	$c \leq 20$	$c \leq 40$	$c \leq 60$	$c \leq 80$	$c \leq 100$	$c \leq 120$	$c \leq 140$	$c \leq 160$
Cumulative frequency	3	11						80

[1]

- (c) On the grid opposite, draw a cumulative frequency curve to represent this data.

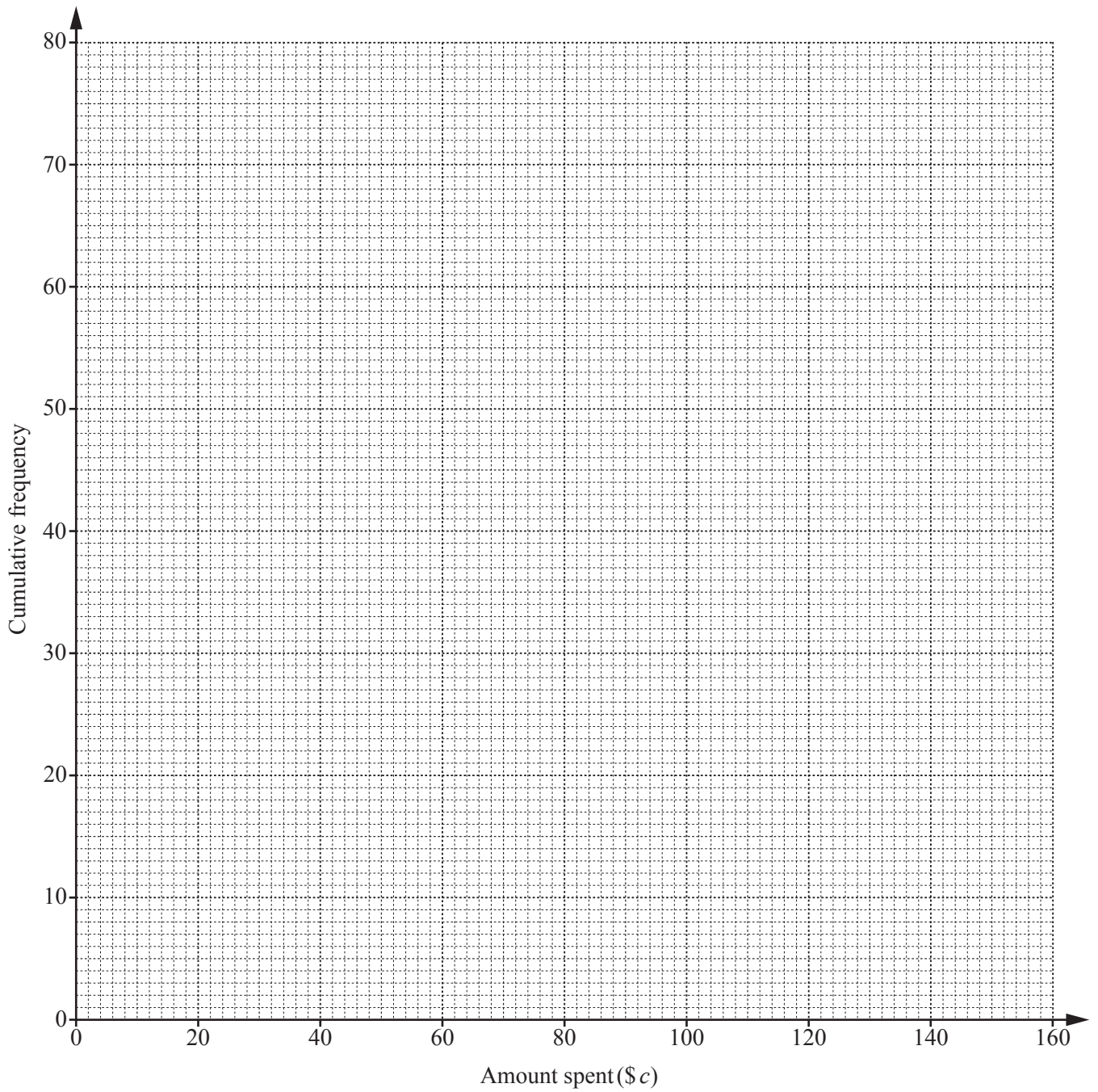
[2]

- (d) (i) Use your graph to estimate the median.

Answer \$ [1]

- (ii) Use your graph to estimate the interquartile range.

Answer \$ [2]



- (e) The number of people who spent more than \$85 last month is the same as the number of people who spent between \$ k and \$85.

Given that k is less than 85, use your graph to estimate the value of k .

Answer $k = \dots\dots\dots$ [3]

Answer Key:

1. S23/qp41/q5(a)(i) (0580)

5(a)(i)(a)	25	1	
5(a)(i)(b)	17 to 18	1	
5(a)(i)(c)	12	2	B1 for 148 seen
5(a)(i)(d)	30	2	B1 for 104 seen

2. W22/qp41/q5(a) (0580)

5(a)(i)	9.4	1	
5(a)(ii)	2.4	2	B1 for [uq =] 10.4 or [lq =] 8 but not as final answer
5(a)(iii)	18	2	B1 for 82 seen

3. W21/qp42/q2(a) (0580)

2(a)(i)	90	1	
2(a)(ii)	68	1	
2(a)(iii)	52	1	FT 120 – <i>their</i> (a)(ii)
2(a)(iv)	20	2	B1 for 60 in working or as answer

4. S21/qp43/q3(b) (0580)

3(b)(i)	14	1	
3(b)(ii)	4	2	B1 for [l.q. =] 11 or [u.q =] 15

5. W20/qp42/q4(a) (0580)

4(a)(i)	25	1	
4(a)(ii)	10 nfw	2	B1 for [lq =] 22 or [uq =] 32
4(a)(iii)	27	1	
4(a)(iv)	6	2	B1 for 114 written
4(b)(i)	27.9 or 27.91 to 27.92 nfw	4	M1 for mid-values M1 for $\sum fx$ where x lies within or on boundary of correct interval M1 dep $\sum fx \div 120$ dep on second M1

6. S21/qp21/q9(a)

9(a)(i)	48	1	
9(a)(ii)	43 to 45	2	B1 for $80 - \frac{60}{100} \times 80$ soi After 0 scored, SC1 for answer 51 to 53
9(a)(iii)	18, 32, 16, 6	2	B1 for 2 or 3 correct

7. W20/qp22/q2

2(a)(i)	Correct cumulative frequency diagram	3	B2 for at least 4 correct points plotted or B1 for 10, 30, 64, 76, 80 soi
2(a)(ii)(a)	Correct reading at CF of 40	1	FT <i>their</i> cumulative frequency diagram
2(a)(ii)(b)	24 to 28	2	FT <i>their</i> cumulative frequency diagram B1FT for correct reading of <i>their</i> LQ or <i>their</i> UQ
2(b)(i)	2.06[25] or 2.063 or $2\frac{1}{16}$	2	M1 for $([0 \times 24] + 1 \times 30 + 2 \times 50 + 3 \times 32 + 4 \times 16 + 5 \times 8) \div 160$
2(b)(ii)	$\frac{24}{160}$ oe	1	
2(b)(iii)	$\frac{29}{848}$ oe	2	M1 for $\frac{30}{160} \times \frac{29}{159}$ After M0, SC1 for answer $\frac{9}{256}$ oe

8. W20/qp21/q2

2(a)(i)(a)	188 to 189	1	
2(a)(i)(b)	37 to 39	2	B1 for 207 to 208 or 169 to 170 seen
2(a)(ii)			M2 for $\frac{60-52}{60} [\times 100]$ oe or $\frac{52}{60} \times 100$ oe or M1 for reading graph at 220 g
2(a)(iii)	B, IQR is smaller oe	1	Strict FT <i>their</i> 2(a)(i)(b)
2(b)	130 or 129.5 to 129.6	3	B1 for correct midpoints soi M1 for $(10 \times 90 + 15 \times 110 + 42 \times 125 + 36 \times 135 + 17 \times 170) \div 120$
2(c)	26.105	3	B1 for 605 g oe or 2.55 kg oe soi M1 for $10p + b$ where $2.4 \leq p \leq 2.6$ and $0.59 \leq b \leq 0.61$

9. W17/qp22/q2

2(a)(i)	12 40 85 107	1	
2(a)(ii)	Correct cumulative frequency curve	2	B1FT for at least 5 correct plots
2(b)(i)	47 to 49	1FT	
2(b)(ii)	28 to 32	2FT	B1 for 63 to 65 or 32 to 35

10. S17/qp22/q11(b)

11(b)(i)	34	1	
11(b)(ii)	4.5	2	B1 for 31.5 to 32.5 <u>and</u> 36 to 37 seen
11(b)(iii)	(28, 0) (32, 15) (36, 45) (40, 60) plotted and points joined	3	B2 for at least 3 correct points plotted or B1 for 2 correct points plotted or (28, 0) (32, 15) (36, 45) and (40, 60) seen

11. S17/qp21/q11

11(a)	75 nfww	3	M2 for $\frac{\sum \text{frequency} \times \text{midvalue}}{80}$ oe or M1 for $\sum fc$
11(b)	25, 46, 64, 73, 78	1	
11(c)	8 points correctly plotted and joined	2	FT increasing curve B1 for at least 6 points correctly plotted
11(d)(i)	74 to 76	1	
11(d)(ii)	36 to 44	2	B1 for 52 to 56 <u>and</u> 92 to 96 seen
11(e)	54 to 62	3	B1 for 27 to 29 M1 for attempt to read at $(80 - 2 \times \text{their } 28)$